

Engine-Native AlignAtt for Decoder-Only LLM MT in an IWSLT 2026 Simultaneous Speech Translation System

Quentin Fuxa

Independent Researcher

quentin.fuxa@gmail.com

Dominik Macháček

Charles University

Institute of Formal and Applied Linguistics (UFAL)
machacek@ufal.mff.cuni.cz

Abstract

We describe our IWSLT 2026 simultaneous speech translation system: a synchronous cascade with Qwen3 forced ASR on the speech side and Gemma-4 E4B-it with AlignAtt on the machine-translation side, evaluated on English-to-German, English-to-Chinese, and English-to-Italian in low-latency and high-latency regimes. The technical obstacle is that the MT model is a decoder-only LLM served through fused, graph-compiled inference, so the cross-attention signal required by standard AlignAtt is neither architectural nor directly observable at runtime. We recover a usable MT-side policy through a deterministic prompt layout, offline selection of translation-specific heads, selective *qk-fast* replay of the draft-to-source attention block, and an engine-native query/key capture path that preserves model outputs bit-identically. On the IWSLT 2026 dev set, the retained cascade reaches a low-latency operating point near two seconds of compute-unaware LongYAAL and trades 1.32–1.53 additional seconds for clear gains in BLEU and XCOMET-XL in the high-latency regime. Head selection remains an end-to-end quality/latency trade-off, and the deployed *qk-fast* path keeps the MT observer cheap enough to stay on the deployed runtime.

1 Introduction

This paper describes the retained IWSLT 2026 submission system for English-to-German, English-to-Chinese, and English-to-Italian. The stack is a synchronous Qwen3 forced ASR + Gemma-4 E4B-it cascade with two operating regimes: one low-latency and one high-latency. Earlier exploration of an all-Gemma “single substrate” variant remains in the appendix, but it is not part of the submitted system.

One deployment choice shapes the rest of the paper: both ASR and MT are served through vLLM so that speech and translation share a single compiled inference stack. On the MT side, that choice

removes the standard AlignAtt observation point twice over. Gemma-4 is decoder-only, so there is no decoder cross-attention to read, and vLLM keeps self-attention inside fused, graph-compiled kernels, so the full attention matrix is not exposed to Python. The main technical contribution is therefore an engine-native AlignAtt realization for decoder-only MT: a deterministic prompt layout to mark the source span, offline head selection, selective replay of the draft-to-source block, and compile-safe query/key capture that preserves model outputs bit-identically.

Section 2 situates the problem, Section 3 describes the submitted system, Section 4 details the MT-side AlignAtt realization, Section 5 reports the results, and the appendix keeps the heavier derivations and exploratory analyses.

2 Background

AlignAtt was introduced for encoder–decoder simultaneous speech translation, where each decoder cross-attention row is normalized over source positions only (Papi et al., 2023). In that setting, source provenance is almost architectural: the online policy can inspect whether a drafted token depends on source frames that are already accessible or still too recent. That is the point of departure for our paper. We keep the same policy intuition, but we have to recover the alignment signal from a very different substrate.

On the MT side, the submitted system uses a decoder-only LLM rather than an encoder–decoder translation model. The source transcript, translation instruction, accepted target prefix, and drafted continuation all live inside one causal prompt, and the deployed runtime fuses attention inside graph-compiled kernels. Prior interpretability work nevertheless shows that multilingual LLMs contain a sparse set of translation-specific heads (Liu et al., 2026). This makes an

AlignAtt-style policy plausible, but only if the source span is made explicit in prompt space and the policy-visible attention rows can be reconstructed from tensors captured on the real inference path.

The evaluation protocol also matters for interpretation. We follow the Simulstream setting of long-form, unsegmented, revision-capable streaming translation (Gaido et al., 2025) and report LongYAAL in both compute-unaware and compute-aware variants (Polák et al., 2025). The submitted cascade itself is synchronous and single-GPU: each chunk triggers ASR first and MT second. This conservative regime cleanly separates policy behavior from scheduler overlap.

Figure 1 summarizes the conceptual shift. The policy intuition remains the same as in encoder-decoder AlignAtt, but the alignment signal must now be recovered from a prompt-structured self-attention substrate rather than read directly from cross-attention.

3 Submitted System Overview

ASR backend. The speech side of the submitted cascade uses Qwen3-ASR with the Qwen3 forced aligner (Qwen Team, 2025). For each chunk boundary, it returns an updated transcript prefix together with per-word end times. We retain this dedicated aligner because it is materially better than the discarded all-Gemma ASR path on boundary accuracy, word error rate, and runtime; the auxiliary comparison is reported in Appendix C.

MT backend. The translation side uses Gemma-4 E4B-it served through graph-compiled vLLM. At step k , Gemma receives the current transcript prefix, the already accepted target prefix, and a fixed translation instruction, then generates a short draft continuation. AlignAtt accepts the longest draft prefix whose alignment signal stays on the currently accessible side of the source frontier.

Streaming loop. The cascade is synchronous and chunk-based. Each chunk first updates the ASR transcript prefix, then launches one MT request. Source words are considered accessible only once their end time is older than a small hold-back margin. MT output is append-only: once a target token is accepted, it is reused as causal prefill for the next MT step and is never retracted.

Submitted operating points. We keep two official submission regimes. The low-latency submission

uses $\Delta_{\text{chunk}} = 850$ ms and the high-latency submission uses $\Delta_{\text{chunk}} = 1500$ ms. Both use the same ASR backend, Gemma-4 MT backbone, and MT-side AlignAtt realization.

4 Engine-Native AlignAtt for LLMs

This section focuses on the MT-side mechanism that makes Gemma-4 a usable AlignAtt backend inside the retained cascade.

4.1 Prompt Layout and Alignment Heads

Standard AlignAtt does not directly apply because a decoder-only LLM has no decoder cross-attention. We therefore recover a source-conditioned view by forcing a deterministic prompt layout:

$$\mathbf{p}^{(k)} = \left[\mathbf{p}^{\text{sys}} \parallel \mathbf{s}^{(k)} \parallel \mathbf{p}^{\text{instr}} \parallel \mathbf{y}_{1:m_k} \parallel \tilde{\mathbf{y}}^{(k)} \right], \quad (1)$$

where $\mathbf{s}^{(k)}$ is the live transcript prefix returned by ASR, $\mathbf{y}_{1:m_k}$ is the already accepted translation prefix, and $\tilde{\mathbf{y}}^{(k)}$ is the draft proposed by Gemma-4 at the current step. The transcript prefix is therefore a contiguous prompt span with a known map $\phi^{(k)}$ from source-word indices to prompt positions.

Not all self-attention heads carry a useful translation signal (Liu et al., 2026). Following Liu et al. (2026), we operationalize this with a two-stage offline calibration procedure under the exact prompt layout above: GPT-5-mini first provides word-level source-target alignments on held-out parallel text, then every Gemma head is scored with Translation Score (TS) on those aligned examples. We retain the top $k = 8$ heads per language pair. This head set is fixed at inference time and is the only part of the policy that depends on offline calibration. Appendix A reports the calibration details, corpus sizes, and per-direction score summaries. The retained MT heads are sparse and concentrated in a limited late-depth region of the backbone rather than spread uniformly; Appendix A visualizes that pattern in Figure 8.

The policy only reads the source slice of each draft attention row, $\mathbf{A}_{t, \phi^{(k)}(s)}^{(\ell, h)}$, but unlike encoder-decoder cross-attention this slice is not source-normalized. Off-source mass on the accepted target prefix, prompt template, and speculative suffix is therefore structural rather than implementation noise. On the 78 French qualitative probes used for our detailed diagnostics, drafted units allocate on average 9% to accessible source tokens, 8%

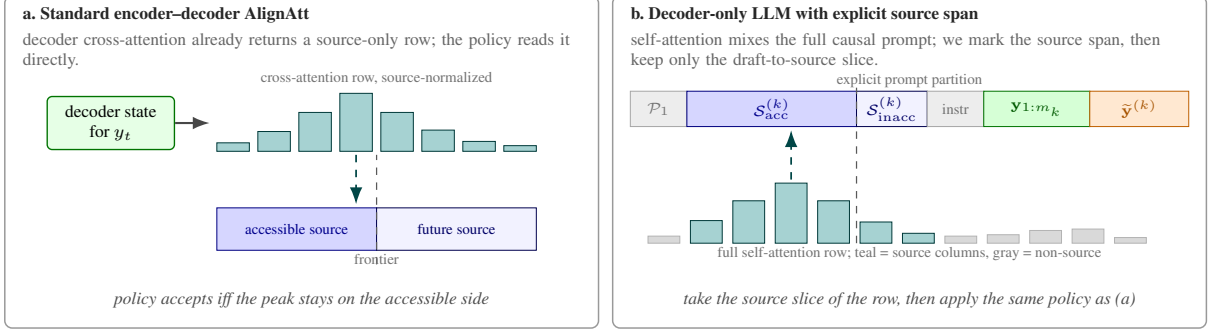


Figure 1: **How AlignAtt changes substrate between encoder-decoder and decoder-only models.** (a) In the original encoder-decoder setting, the decoder already exposes a source-only cross-attention row, so the policy can gate tokens directly against the accessible source frontier: if the peak of the row falls on the accessible side, the draft token is accepted. (b) In our decoder-only MT setting, source and target history share one causal prompt. We therefore mark the source span explicitly in prompt space, isolate translation-specific heads, and reconstruct only the draft-to-source slice needed by the policy; the decision rule is otherwise the same as in (a).

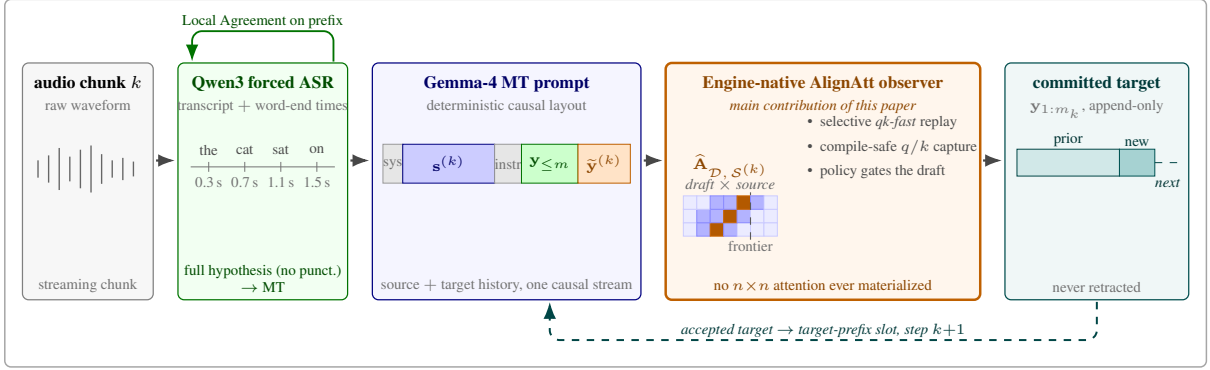


Figure 2: **Submitted cascade, one chunk-synchronous step.** Each chunk first updates the source prefix with Qwen3 forced ASR, then runs one Gemma-4 MT step. The MT prompt keeps the ASR transcript explicit inside the decoder-only causal layout, so the engine-native observer (*main contribution of this paper*) can capture the selected heads’ queries and keys on the deployed vLLM path and reconstruct only the draft-to-source block $\hat{A}_{\mathcal{D},S^{(k)}}$ used by the acceptance policy. Accepted target tokens are appended to $y_{1:m_k}$ and become the target-prefix slot of the next step’s prompt (dashed teal feedback).

to inaccessible source tokens, 81% to non-source prompt positions, and 2% to the speculative suffix; among units that the policy actually accepts, the inaccessible-source share drops from 8% to 1%.

4.2 Selective Reconstruction for the Policy

The policy never needs the full $n \times n$ self-attention matrix. It only needs draft rows against source columns. For selected heads (ℓ, h) , draft positions t , and source positions s , we reconstruct

$$\hat{A}_{t,s}^{(\ell,h)} = \frac{\exp\left(q_t^{(\ell,h)} \cdot k_{\phi^{(k)}(s)}^{(\ell,h)}\right)}{\sum_j \exp\left(q_t^{(\ell,h)} \cdot k_j^{(\ell,h)} + m_{t,j}\right)}, \quad (2)$$

where the denominator still ranges over the entire prompt. We call this replay path *qk-fast*: it is exact up to floating-point reassociation relative to

the fused forward and only reconstructs the policy-visible block.

From these per-head rows we build two source-side provenance features for each drafted token: accessible-source mass and inaccessible-source mass,

$$\pi_t^{\text{acc}} = \sum_{s \in S_{\text{acc}}^{(k)}} \bar{\mathbf{r}}_t(s), \quad (3)$$

$$\pi_t^{\text{inacc}} = \sum_{s \in S_{\text{inacc}}^{(k)}} \bar{\mathbf{r}}_t(s), \quad (4)$$

where $\bar{\mathbf{r}}_t$ is the head-averaged reconstructed row. We additionally normalize per-head rows online with prefix Welford statistics and apply a short median filter along the source axis before taking the source-side argmax. This stabilizes the alignment peak without changing the underlying replayed

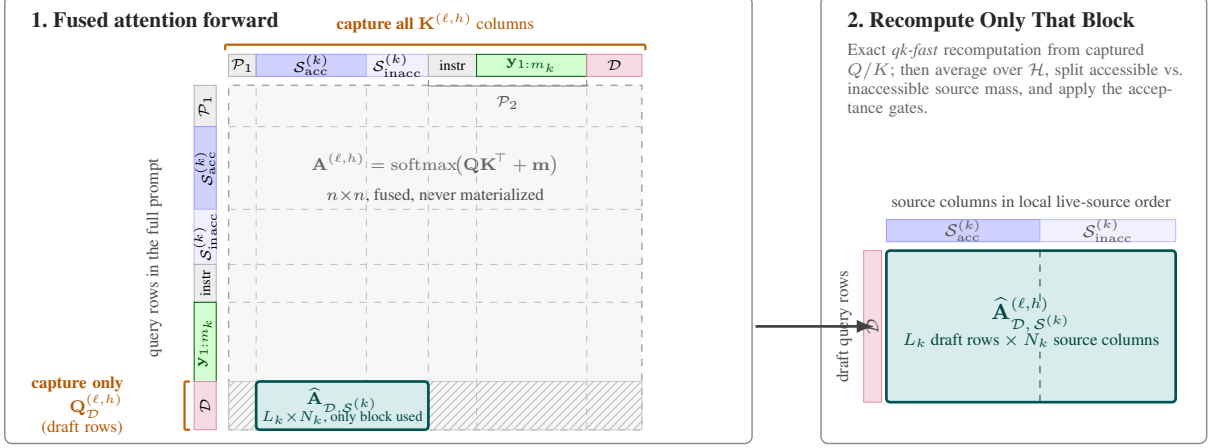


Figure 3: **Selective reconstruction with engine-native capture.** The $n \times n$ attention matrix $\mathbf{A}^{(\ell,h)}$ is executed entirely inside the fused attention kernel, so the full $n \times n$ matrix is never materialized. For each AlignAtt head $(\ell, h) \in \mathcal{H}$, the observer copies $\mathbf{K}^{(\ell,h)}$ for all positions and $\mathbf{Q}^{(\ell,h)}$ only for the draft rows $\mathcal{D}$ into fixed-shape buffers. The green band $y_{1:m_k}$ inside $\mathcal{P}_2$ marks the committed target prefix that grows across chunks; hatched cells are never reconstructed. After the forward, Eq. (2) recomputes exactly only the draft-to-source block $\hat{\mathbf{A}}_{\mathcal{D},S^{(k)}}^{(\ell,h)}$, which is the sole object consumed by the source-side provenance scores of Eqs. (3)–(4) and by the acceptance policy of this section.

rows.

4.3 AlignAtt Acceptance Policy

The submitted MT policy is a first-failure scan over drafted tokens. Let $\hat{s}_t$ be the source-side argmax of the filtered row for draft token t , and let $N_{\text{acc}}^{(k)}$ be the number of source words whose ASR end time is already on the accessible side of the frontier. The scan stops when one of three conditions first fires:

$$\text{SOURCE-FRONTIER : } \hat{s}_t \geq N_{\text{acc}}^{(k)} + b, \quad (5)$$

$$\text{ARGMAX-MASS-WEAK : } \bar{\mathbf{r}}_t(\hat{s}_t) < \tau_{\text{argmax}}, \quad (6)$$

$$\text{PROVENANCE-WEAK : } \pi_t^{\text{acc}} < \tau_{\text{src}}. \quad (7)$$

Figure 4 makes the decision path explicit. One branch averages the retained heads and sums accessible-source mass to obtain π_t^{acc} . The other normalizes and smooths the same replayed rows before taking the source-side argmax $\hat{s}_t$. The gate accepts token t iff all three conditions above pass; a left-to-right first-failure scan then emits the longest accepting draft prefix and rounds back to the nearest stability-unit boundary so partial subword fragments are never emitted.

4.4 Engine-Native Capture under Compiled Inference

Equation (2) is only useful if the deployed runtime exposes the exact queries and keys consumed

by attention. Under fused, graph-compiled vLLM, those tensors never appear as ordinary Python-visible objects. We therefore install a per-layer observer before compilation, copy prompt keys and draft-row queries into fixed-shape buffers, and route capture through a custom-op-style call that survives graph lowering. A zero-valued sentinel dependency keeps the observer in the transitive fan-in of the graph output:

$$\mathbf{o}'_\ell = \mathbf{o}_\ell + \Phi(\ell, \text{pos}, \mathbf{Q}^{(\ell)}, \mathbf{K}^{(\ell)}), \quad \Phi \equiv \mathbf{0}. \quad (8)$$

Figure 5 separates the capture path into three stages. At compile time, we patch the attention forward and install fixed-shape observer slots before graph capture. During each compiled forward, the live observer records prompt keys and draft-row queries for the selected heads while returning $\Phi \equiv \mathbf{0}$ back into the attention output. After the forward, a short Python-side replay reconstructs only the selective draft-to-source block consumed by the policy and by the parity suite. The sentinel term is therefore only a liveness device: without the additive zero, graph lowering dead-code-eliminates the observer path and no usable Q/K buffers survive on the deployed runtime. Appendix B keeps the replay equations, parity measurements, and qualitative probe.

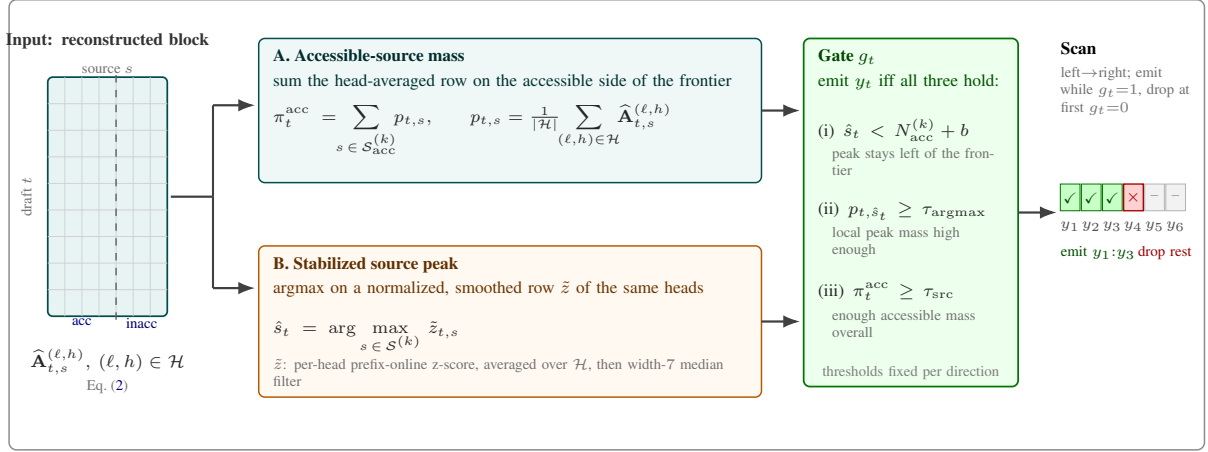


Figure 4: **From the reconstructed block to an acceptance decision.** The selective *qk-fast* reconstruction produces $\hat{\mathbf{A}}_{t,s}^{(\ell,h)}$ for $(\ell, h) \in \mathcal{H}$, with draft positions t against source positions s (left). The policy reads it through two parallel aggregations. *Branch A* averages the selected heads into a row $p_{t,\cdot}$, and sums its mass on the accessible side of the frontier, giving the provenance score π_t^{acc} . *Branch B* returns the peak $\hat{s}_t$ of a stabilized row $\tilde{z}_{t,\cdot}$, where $\tilde{z}$ is obtained by z-scoring each head with prefix-online Welford moments, averaging over $\mathcal{H}$, and applying a width-7 median filter along the source axis. The gate g_t accepts draft token t iff the peak stays on the accessible side of the frontier, the peak mass exceeds τ_{argmax} , and the accessible-source mass exceeds τ_{src} . A left-to-right scan emits the longest run of accepting tokens and drops the remainder at the first failure.

5 Results

5.1 Experimental Setup

All experiments run on a single NVIDIA A40. The MT backend uses Gemma-4 E4B-it in bfloat16 through graph-compiled vLLM; the speech side uses Qwen3-ASR with forced alignment. Evaluation uses the official 21-clip IWSLT 2026 MCIF dev set (~ 2.1 hours total, long academic talks) resegmented by OmniSTEval v0.1.7. We report BLEU and chrF against the dev references, XCOMET-XL for translation adequacy (Guerreiro et al., 2024), and compute-unaware / compute-aware LongYAAL for latency (Polák et al., 2025). The streaming loop follows the Simulstream setting of unsegmented long-form audio with revision-capable evaluation (Gaido et al., 2025), but our submitted cascade itself is synchronous: ASR and MT are serialized on one GPU so that policy effects are not confounded with scheduler overlap.

For the auxiliary runtime study in Section 5.4, we compare three MT seams on a fixed suite of 16 text-only prompts: a minimal Transformers eager reference, a Transformers SDPA – *qk-fast* replay reference, and the deployed vLLM *qk-fast* path. Each seam uses greedy decoding with the same head set, one warmup pass, and repeated hot runs in isolated subprocesses.

Pair	Regime	BLEU	chrF	XCOMET-XL	LongYAAL (s)	
					CU	CA
en→de	low	28.76	62.1	0.875	2.00	1.63
	high	32.63	64.2	0.902	3.53	3.14
en→zh	low	36.01	35.0	0.743	1.95	1.77
	high	39.86	37.8	0.778	3.27	3.09
en→it	low	40.10	68.0	0.805	1.98	1.62
	high	44.46	70.1	0.841	3.48	3.10

Table 1: **Submitted cascade on the IWSLT 2026 dev set** (21 clips, OmniSTEval long-form resegmentation). Low rows use the submitted setting with $\Delta_{\text{chunk}} = 850$ ms. High rows use the submitted setting with $\Delta_{\text{chunk}} = 1500$ ms.

5.2 Submitted System Results

Table 1 reports the submitted operating points. The pattern is consistent across EN→DE, EN→ZH, and EN→IT: the low-latency regime stays near a 2 s CU-LongYAAL point, while the high-latency regime moves to a clearly stronger quality point at the expected latency cost. We therefore read Table 1 as evidence that the MT-side AlignAtt realization supports both a stable low-latency operating point and a higher-quality regime on the same decoder-only Gemma backend.

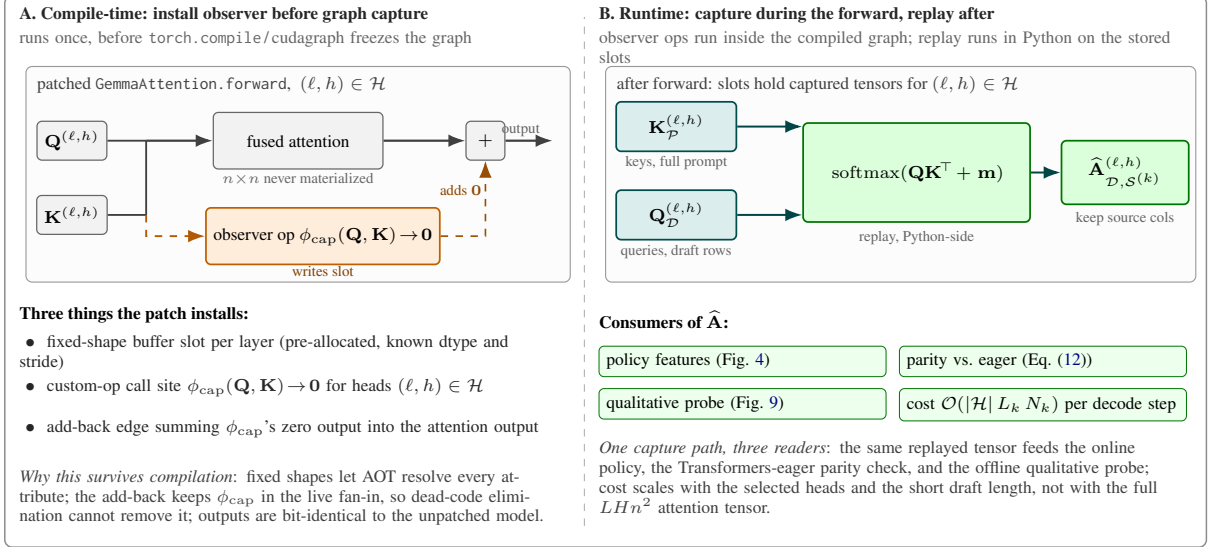


Figure 5: **Compile-safe observer lifecycle on the deployed vLLM path.** *Left (A), once, before graph capture:* the forward of `GemmaAttention` is patched so that for each selected head $(\ell, h) \in \mathcal{H}$ the query and key tensors also pass through an observer custom op ϕ_{cap} (dashed orange side path). The op writes the selected slices into fixed-shape, pre-allocated slots and returns a zero tensor that is added back into the attention output; this keeps the op in the live fan-in of the graph so the compiler cannot dead-code-eliminate it, while the numerical output of the layer is unchanged. *Right (B), every forward:* after the compiled forward finishes, the observer slots hold $K_P^{(\ell, h)}$ for the full prompt and $Q_D^{(\ell, h)}$ for the draft rows; a short Python-side replay computes QK^T for the draft against the full prompt, applies the causal mask m , takes the softmax, and keeps the source columns to yield the selective block $\hat{A}_{D, S^{(k)}}^{(\ell, h)}$ that feeds the policy, the parity check against an uncompiled Transformers reference, and the qualitative probe.

5.3 Head Filtering Matters End-to-End

Table 2 is an auxiliary EN→DE rerun included only to isolate the MT head-set effect under the maintained *qk-fast* runtime. End-to-end quality is nearly equivalent, but the latency split is informative. CU-LongYAAL rises by +100.3 ms, which likely means the all-head observer is slightly less precise and therefore a bit more generous on tokens whose support extends beyond the currently available source prefix. CA-LongYAAL rises more strongly, by +179.5 ms, because replaying all 336 heads forces the runtime to reconstruct much more attention state at each MT step. We therefore read the auxiliary comparison as evidence that a small retained head set captures most of the useful MT alignment signal at much lower runtime cost. Appendix A gives the held-out calibration numbers behind the retained head sets.

5.4 Runtime Cost of the MT Observer

The MT observer must be cheap enough to justify deployment. Figure 6 shows that the engine-native vLLM replay path is much cheaper than the minimal eager seam we use as an inspection reference: median cost drops from 63.7 to 25.4

Setting	XCOMET-XL $\uparrow$	CU (s) $\downarrow$	CA (s) $\downarrow$
Top-8 heads	0.879	1.96	1.65
All 336 heads	0.885	2.06	1.83

Table 2: **Auxiliary EN→DE MT head-set comparison at $\Delta_{\text{chunk}} = 1100$ ms.** Both rows use the same maintained runtime; only the MT head set changes. End-to-end quality is nearly equivalent, but all-head replay increases CU and especially CA because the observer becomes less selective while the runtime must reconstruct much more attention at each MT step.

ms/token, while the Transformers SDPA – *qk-fast* replay remains close to the eager baseline at 59.0 ms/token. This confirms that the MT contribution is not merely inspectable in principle; it is practical inside the compiled runtime we actually deploy.

5.5 Coordination Regimes and Compute-Aware Latency

Our cascade loads the ASR and MT models as two separate blocking `vllm.LLM` instances on a single GPU, which serializes every audio chunk into an ASR decode followed by an MT decode

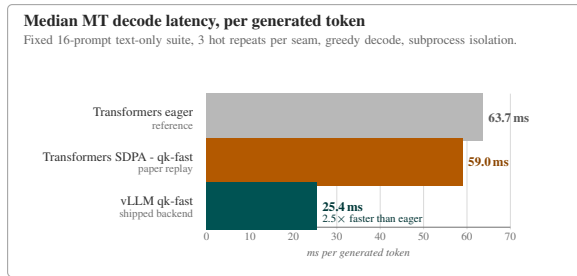


Figure 6: **Inference-time comparison of MT capture seams.** Median latency per generated token on a fixed 16-prompt text-only suite, for a minimal Transformers eager reference, a Transformers SDPA – *qk-fast* reference that reconstructs source rows from captured layer inputs, and the engine-native vLLM *qk-fast* path used by the shipped backend. Each seam uses one unmeasured warmup pass followed by three hot repeats, greedy decode, a fixed AlignAtt head set, and subprocess isolation; per-prompt results and divergences are stored in the generated JSON artifact.

(Fig. 7(c)). This conservative regime keeps policy effects separate from scheduler overlap. Alternative pipelined or fully asynchronous regimes would mainly reduce compute-aware LongYAAL at fixed compute-unaware latency, but would require a non-blocking scheduler and a more careful treatment of stale MT requests. We leave that scheduler axis to future work.

6 Conclusion

The retained IWSLT 2026 system is a Qwen3 forced ASR + Gemma-4 MT cascade, and its key technical ingredient is an engine-native realization of AlignAtt for decoder-only MT under compiled inference. Deterministic prompt layout, offline head selection, selective *qk-fast* replay, and compile-safe query/key capture make the MT policy usable without leaving the deployed runtime. On the dev set, this yields a low-latency operating point near 2 s CU-LongYAAL and a high-latency regime that is roughly 1.5 s slower but clearly stronger in BLEU and XCOMET-XL.

References

- Marco Gaido, Sara Papi, Mauro Cettolo, Matteo Negri, and Luisa Bentivogli. 2025. Simulstream: Open-source toolkit for evaluation and demonstration of streaming speech-to-text translation systems. ArXiv preprint arXiv:2512.17648.
- Nuno M. Guerreiro, Ricardo Rei, Daan van Stigt, Luisa Coheur, Pierre Colombo, and André F. T. Martins. 2024. xCOMET: Transparent machine transla-

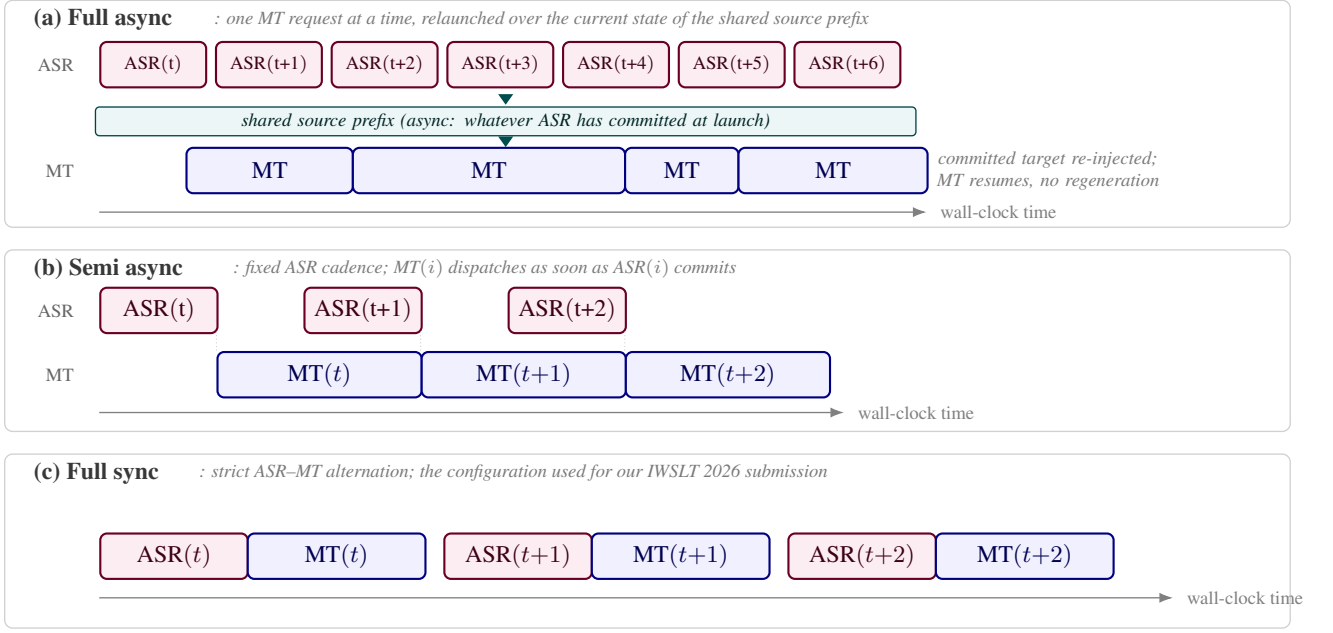
tion evaluation through fine-grained error detection. *Transactions of the Association for Computational Linguistics*, 12:979–995.

Binbin Liu, Wenhan Han, Feng Chen, Yifan Zhang, Ping Guo, Haobin Lin, Bingni Zhang, Taifeng Wang, and Yin Zheng. 2026. Token alignment heads: Unveiling attention’s role in LLM multilingual translation. ICLR 2026 poster, OpenReview.

Sara Papi, Marco Turchi, and Matteo Negri. 2023. Alignatt: Using attention-based audio-translation alignments as a guide for simultaneous speech translation. In *Proceedings of Interspeech 2023*.

Peter Polák, Sara Papi, Luisa Bentivogli, and Ondřej Bojar. 2025. Better late than never: Meta-evaluation of latency metrics for simultaneous speech-to-text translation. ArXiv preprint arXiv:2509.17349.

Qwen Team. 2025. Qwen3-ASR: A multilingual speech recognition model with forced-alignment timestamp supervision. Technical report, Alibaba Cloud.



Regimes (a) and (b) require a non-blocking scheduler (e.g. vLLM AsyncLLMEngine) with request abort on stale MT decodes; our current implementation is (c) with two blocking LLM instances sharing a single GPU. The gaps between ASR chunks in (a) and (b) reflect an ASR real-time factor below 1. This design axis mainly affects *compute-aware* latency; our central submitted comparisons remain on the serialized regime in (c).

Figure 7: Three coordination regimes for ASR and MT sharing one GPU. The submitted system uses the strict ASR–MT alternation in (c). In full async mode (a), successive MT launches need not correspond to comparable source-time increments: a long MT decode can absorb several incoming ASR updates, while the next relaunch may cover much less newly available audio. That makes history-dependent policies such as local agreement less natural to calibrate there than an attention-only policy such as AlignAtt, which scores each draft directly against the currently accessible source prefix.

A Additional MT Alignment-Head Diagnostics

This appendix keeps only the MT-side head diagnostics omitted from Section 4. Figure 8 shows the architecture-aware score map behind the retained MT head set: useful heads are sparse, clustered late in the backbone, and overlap the late shared-KV region where observer cost is governed by KV ownership rather than raw query-head count. Table 3 then reports the held-out calibration gap between the retained top-8 heads and uniform all-head averaging.

B Observer Replay and Qualitative Diagnostics

This appendix keeps only the details omitted from Section 4: the prompt/suffix replay equations, one qualitative probe, and the numerical parity measurements on the engine-native path.

Prompt-space replay details. After capture, we replay prompt-side and suffix-side logits separately from the stored buffers. Writing $\mathbf{B}_*^{(\ell,h)} := \mathbf{B}_*^{(\ell)}[h]$ for head-specific captured tensors, we

compute

$$\mathbf{P}^{(\ell,h)} = \mathbf{B}_{\text{dQ}}^{(\ell,h)} (\mathbf{B}_{\text{pK}}^{(\ell,h)})^\top, \quad (9)$$

$$\mathbf{S}^{(\ell,h)} = \mathbf{B}_{\text{dQ}}^{(\ell,h)} (\mathbf{B}_{\text{dK}}^{(\ell,h)})^\top + \mathbf{m}_{\text{causal}}, \quad (10)$$

$$\tilde{\mathbf{A}}^{(\ell,h)} = \text{softmax}[\mathbf{P}^{(\ell,h)}, \mathbf{S}^{(\ell,h)}]. \quad (11)$$

Restricting $\tilde{\mathbf{A}}^{(\ell,h)}$ to the source columns $\phi^{(k)}(s)$ recovers the policy-visible block of Eq. (2). The replay cost therefore scales with the selected head set and the short draft length, not with the full LHn^2 attention tensor.

Qualitative reconstruction example. Figure 9 shows the full reconstruction stack on a text-only MT probe: the exact prompt partition seen by the decoder, word-level aggregation of the replayed

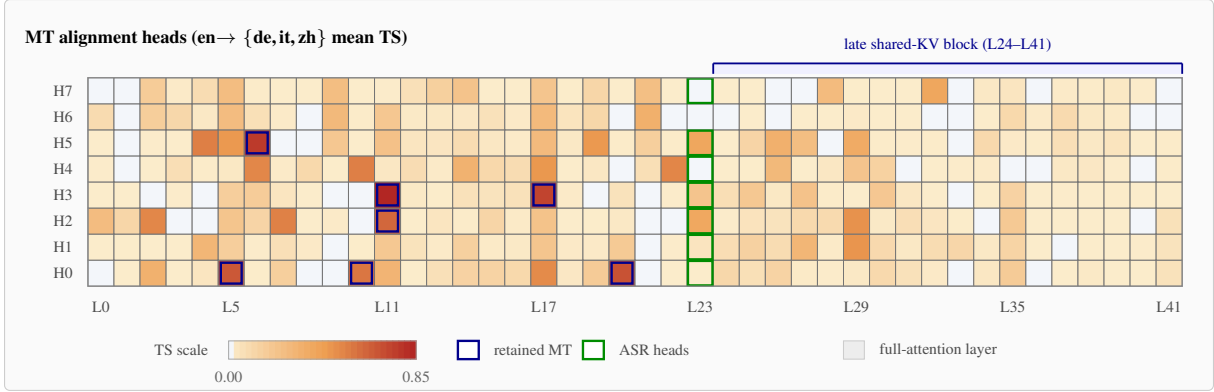


Figure 8: **Architecture-aware view of retained MT alignment heads.** The heatmap covers Gemma’s 42 layers (horizontal axis) and 8 query heads (vertical axis); color encodes mean token-alignment score TS on a shared $[0, \max]$ scale, so empty cells read as low rather than “below threshold”. Scores are averaged over $\text{en} \rightarrow \{\text{de}, \text{it}, \text{zh}\}$. Blue outlines mark the retained shared-kernel MT head set $\mathcal{H}_{\text{MT}}$; green outlines mark the ASR alignment heads from the discarded single-substrate path; grey vertical stripes mark full-attention layers; the blue bracket delimits the late shared-KV block, where observer cost follows KV-group ownership rather than raw query-head count. The useful MT heads are sparse and concentrated in the late part of the backbone, while the limited overlap with the ASR set highlights role-specific specialization.

Pair	Top-8 TS	All-336 TS	Gain	Aligned tokens
EN→DE	90.40	68.49	+21.91	11,209
EN→ZH	93.48	65.79	+27.70	7,582
EN→IT	91.90	67.42	+24.49	12,056

Table 3: **MT head-set filtering ablation on held-out word-aligned dev examples.** To make the all-head baseline maximally charitable, we average the candidate head set first and then restrict the argmax to the source-prompt token zone before scoring against gold aligned source tokens. Scores are reported in points ($100 \times \text{TS}$). Even under this source-only argmax, the per-direction retained top-8 heads still preserve a markedly stronger alignment signal than uniform all-head averaging.

rows for readability, and the four-way provenance accounting for every drafted word.

The green band in Figure 9 is previously committed target text reused as causal context; it is not itself the object of the gate. The orange words are the current draft scanned left-to-right by the policy. We allow $\hat{s}_t$ to move backward within a live draft when the decoder revisits earlier source material under reordering, which is why the residual prompt and suffix masses remain explicit in the decoder-only formulation.

Numerical parity. Let $\mathbf{A}^{(\text{TF})}$ denote the reference attention tensor produced by an uncompiled Transformers forward on the identical prompt. On

a curated parity set, we observe

$$\begin{aligned} \|\Delta \mathbf{A}\|_{\infty} &\leq 1.2 \times 10^{-2}, \\ \|\Delta \mathbf{A}\|_1 / (nL_k) &\leq 4 \times 10^{-4}, \end{aligned} \quad (12)$$

with bit-identical acceptance decisions between the eager and compiled paths. The observer is therefore not just fast enough to keep, but faithful enough to support controlled MT ablations.

C Additional ASR Analyses

This appendix collects the all-Gemma ASR analyses that informed the final system choice but are no longer central to the retained submission narrative.

Gemma AlignAtt ASR is feasible but not competitive. We verified that the same AlignAtt toolkit can be made to run on the ASR side, but the resulting Gemma-ASR path is not competitive enough for submission. The limiting factor is poor Gemma-4 ASR quality itself, and this weakness appears to be shared by the current multi-modal family rather than by one particular size variant: the same qualitative limitation is present for both E2B and E4B. On the diagnostic clip ccpXHNfaoY (360 s), Qwen3 forced clearly dominates the exploratory Gemma AlignAtt ASR path: WER drops from 0.269 to 0.113, CER from 0.122 to 0.046, mean absolute boundary lag from 0.403 s to 0.145 s, load time from 123.0 s to 63.7 s, and wall-clock RTF from 0.628 to 0.306. We therefore

Word-level selective reconstruction on a live MT draft

target=French • gate=alignatt:source_frontier • blocked at source word 13 (*fast*)

system

source

accepted target prefix

draft

Translate to French.	we separately benchmark the eager path before we compare	it with qk fast	nous	évaluons séparément le chemin eager avant de le comparer à qk fast
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Draft-to-source attention (word level)

Per-word provenance

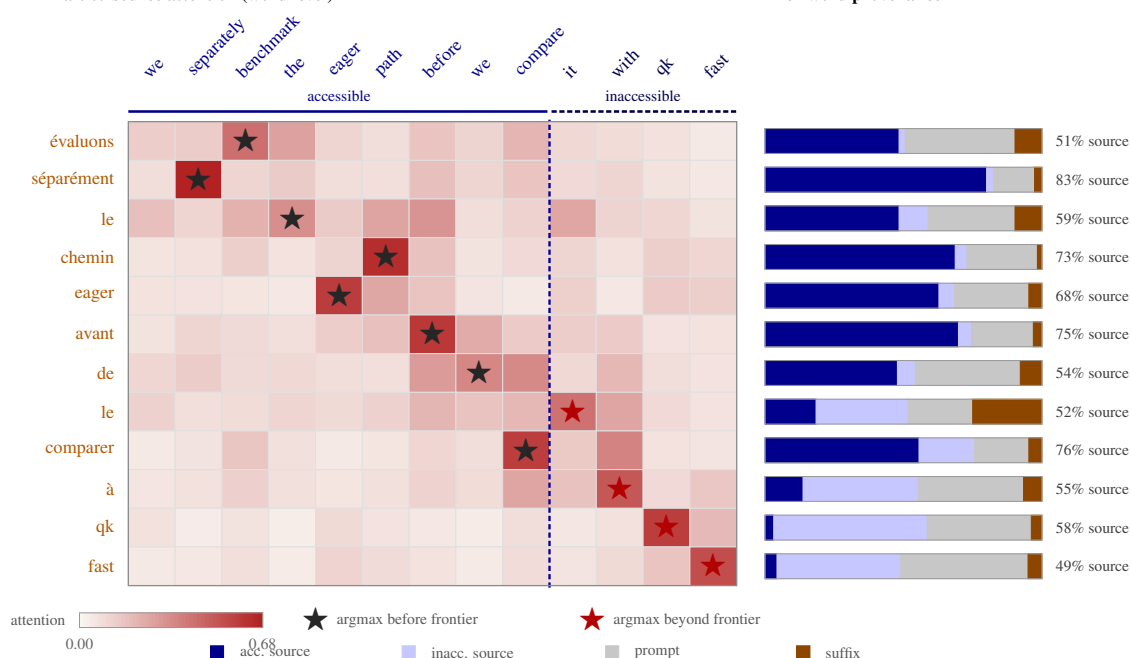


Figure 9: **Word-level selective reconstruction on a live MT draft.** *Top ribbon:* prompt partition into system instruction, live source, accepted target prefix, and current draft. *Left panel:* reconstructed draft-to-source attention from the selected AlignAtt heads, aggregated to words and split by the dashed accessibility frontier; black stars stay on the accessible side, while the first red star marks the SOURCE-FRONTIER failure. *Right panel:* each drafted word is decomposed into accessible-source, inaccessible-source, prompt, and suffix mass, with the numeric label giving the total source share.

keep Gemma AlignAtt ASR only as an exploratory result and retain Qwen3 forced ASR for the submitted system. If a future multimodal LLM closes that ASR gap, the same overall design could support a double-AlignAtt cascade in which AlignAtt governs both the ASR stage and the MT stage.